COUNTING METHODS

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Basic Counting Principles

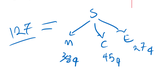
- Counting principle is all about choices we might make given many possibilities.
- It is used to find the number of possible outcomes.
- It provides a basis of computing probabilities of discrete events.

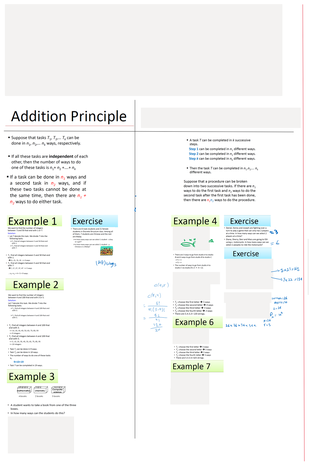
Some sample of counting problems:

- Problem 1 - How many ways are there to seat n couples at a round table, such that each couple sits together?
- Problem 2- How many ways are there to express a positive integer n as a sum of positive integers?
- Problem 3 - There are three boxes containing books. The first box contains 15 mathematics books by different authors, the second box contains 12 chemistry books by different authors, and the third box contains 10 computer science books by different authors. A student wants to take a book from one of the three boxes. In how many ways can the student do this?

Problem 4 - The department will award a free computer to either a CS student or a CS professor. How many different choices are there, if there are 530 students and 15 professors?

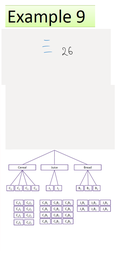
Problem 5 - A scholarship is available, and the student to receive this scholarship must be chosen from the Mathematics, Computer Science, or the Engineering Department. How many different choices are there for this student if there are 38 qualified students from the Mathematics Department, 45 qualified students from the Computer Science Department and 27 qualified students from the Engineering Department?

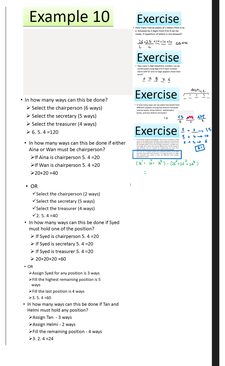




- The counting problem that we have considered so far involved either the addition principle or the multiplication principle.
- Sometimes, however, we need to use both of these counting principles to solve a particular problem.



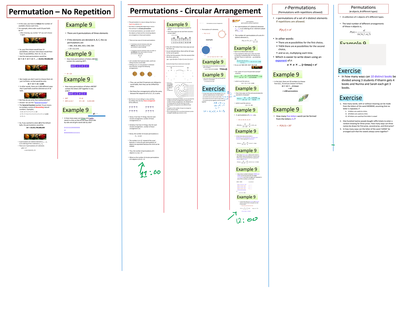




Permutation & Combination

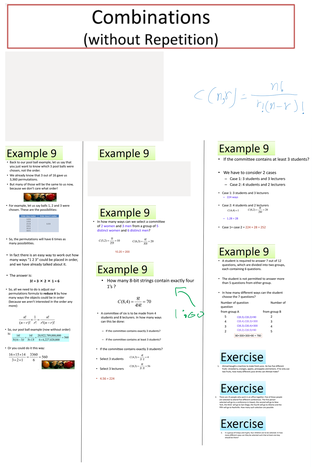
Permutations

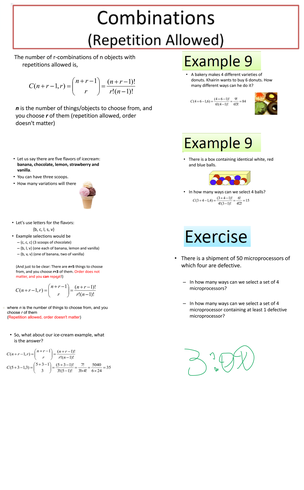




Combinations

- There are also two types of combinations (remember the order does **not** matter now);
- **Repetition is Allowed**: such as coins in your pocket (5,5,5,10,10)
- **No Repetition**: such as combination of subjects to enroll (Maths., English, Music)





Summary

Which formula to use?

	Order Matters (Permutations)	Order Does Not Matter (Combinations)
Repetition is allowed	$P_n = n^r$	$C(n+r-1,r) = \frac{(n+r-1)!}{r!(n-1)!}$
Repetition is not allowed	$P(n,r) = \frac{n!}{(n-r)!}$	$C(n,r) = \frac{n!}{r!(n-r)!}$